Prayer Times and Adhan Mobile Application

Solution Architecture and Delivery Roadmap

Document	Value
Version	1.0
Status	Architecture baseline for product design and delivery planning
Date	24 September 2026
Primary audiences	Business, product, UX, architecture, mobile, backend, QA, security and operations
Working engine name	Geo Salah Engine

Decision summary

Build an offline-first Flutter application whose defining assets are an explainable prayer-time engine and a native alarm layer. Core calculations run on device. Small signed data packages keep country profiles, time-zone rules, calibration evidence and optional mosque coordinates current. The service remains free, requires no login, carries no advertising and has no paid runtime dependency on a competing prayer application or mosque partnership.

Document purpose and reading guide

This document converts the approved product direction into an implementable architecture and delivery plan. It explains the product in business terms, then defines the engineering design, data contracts, controls, quality gates and release sequence needed to build it. The main architectural conclusion is that prayer calculation and alarm delivery must remain independent concerns: the first produces a traceable time, while the second makes a best-effort platform-compliant commitment to alert the user at that time.

Reader	What this document provides
Business sponsors	Value proposition, scope boundaries, success measures, risks and release outcomes
Product management	Capability priorities, release gates, sprint scope and acceptance criteria
UX and UI	Core journeys, status language, confidence disclosure and platform-specific constraints
Solution architecture	Logical components, deployment topology, data ownership and update flows
Mobile and backend engineering	Calculation pipeline, native alarm boundaries, storage, APIs and versioning
QA and accuracy lab	Reference corpus, discrepancy classification, device matrix and release thresholds
Security and privacy	Threat controls, minimum data collection, signing, audit and retention

Document map

Part	Sections	Purpose
A	1 to 4	Product vision, scope, experience and accuracy contract
B	5 to 13	Geo Salah calculation, profiles, calibration, location, time-zone, mosque and Iqama design
C	14 to 20	Qibla, alarm reliability, offline operation, privacy, architecture and data model
D	21 to 27	APIs, dependencies, updates, evaluation, observability, NFRs and testing
E	28 to 33	CI CD, releases, sprint plan, governance, risks and decisions
Appendices	A to E	Schemas, sample evidence, glossary and references

Assumptions and boundaries

- The first production clients are native mobile applications delivered from one Flutter codebase, with Swift and Kotlin modules for platform-critical alarms, widgets and wearables.
- The application is free at the point of use, contains no advertising, does not sell location data and does not require an account for its core experience.

- The engine calculates prayer entry times from exact coordinates. Administrative-area times may be displayed only when a Country Profile explicitly requires an authority convention.
- A selected mosque receives an independent coordinate calculation. Geographic proximity alone does not reveal the mosque's actual clock, loudspeaker delay or congregational schedule.
- Iqama is labeled Estimated unless it comes from a traceable, current and approved source. User observations never silently become verified mosque data.
- No service-level promise can override mobile operating-system controls. The product reports readiness, requests the right permissions and detects degradation honestly.

1 Product vision and governing principles

The product will become the trusted place to answer three questions: when does the next prayer begin here, why is that the displayed time, and will this phone alert me reliably? The product will earn trust through transparent calculation and operational evidence rather than by accumulating unrelated content.

Principle	Business meaning	Engineering consequence
Accuracy before breadth	Prayer time is the primary product promise	Calculation changes require evidence, review and regression tests
Explain every result	Users can understand where a time came from	Every result stores inputs, rule versions and adjustment lineage
Local first	Core value remains available without a connection	On-device calculation, local schedule cache and bundled baseline data
Reliability is observable	The app does not claim an alarm is ready when the OS may suppress it	Permission, audio, battery and schedule health checks
Privacy by default	Location supports the calculation and is not a commercial asset	No account, local storage, coarse telemetry and opt-in diagnostics
Clear confidence	Calculated, calibrated, verified and estimated values are visibly different	Typed provenance and consistent UX labels
Controlled independence	The product is not blocked by a competitor or paid data provider	Internal algorithms, open datasets, cached packages and replaceable adapters
Platform honesty	iOS and Android feel consistent without hiding OS differences	Shared UI and domain logic; native implementations for alarms and system surfaces

2 Business outcomes and success measures

Product success should be measured through accuracy, delivered-alert reliability, trust and repeat use. Downloads alone can hide a technically unreliable product. The first year should use the following scorecard, with final thresholds approved after the evaluation lab establishes realistic baselines.

Outcome	Measure	Initial release target
Authority alignment	Absolute difference against approved authority reference in benchmark cases	At least 95 percent within one minute; no unexplained case above two minutes
Calculation determinism	Same signed input package produces the same result across supported clients	100 percent across conformance corpus
Alarm readiness	Enabled users passing all device readiness checks	At least 90 percent after guided remediation
Alarm delivery	Scheduled alerts observed within the platform tolerance in device tests	At least 99 percent in controlled device lab

Outcome	Measure	Initial release target
Offline continuity	Days of correct schedules after last successful update	At least 30 days, subject to unchanged local time-zone law
Trust	Users opening Accuracy Passport or completing Athan Test	Instrumented without user identity; baseline in beta
Privacy	Core flows possible without account or advertising identifier	100 percent
Quality	Severity 1 defects in calculation or alert delivery at release	Zero open

3 Product scope and user experience

3.1 Primary journeys

1. First run explains why location is useful, offers precise location or manual selection, resolves the governing Country Profile and immediately calculates today's schedule.
2. The home screen shows the next prayer, countdown, the complete day, the active source mode and a compact verification state.
3. A user can open Accuracy Passport to inspect coordinates, elevation source, time zone, calculation rule, calibration version and final adjustment path.
4. A user enables Athan, completes the platform permission flow, runs Athan Test and sees a readiness result before relying on the next alert.
5. A user may select a mosque. The app calculates prayer entry at that mosque coordinate and keeps mosque clock or Iqama information clearly separate.
6. Travel Mode detects a meaningful location or time-zone change, asks before changing the active schedule when appropriate, and recalculates all future alerts.

3.2 Core status language

Label	Meaning	Permitted use
Calculated	Produced by the local engine from stated inputs and rules	Exact-location times and mosque-coordinate prayer entry
Authority aligned	Calculation is within the approved profile tolerance for its evidence set	Only after a published evaluation result
Verified	Received from a traceable approved source with freshness and integrity metadata	Official timetable or mosque data when provenance is valid
Estimated	A model or rule provides a likely range; it is not a confirmed schedule	Iqama estimates and uncertain venue data
Needs attention	A permission, stale package or system condition may reduce accuracy or delivery	Athan Health and data freshness states

3.3 UX requirements for confidence

- Show minutes by default; expose seconds only in expert views or test tools because atmospheric and authority conventions do not justify second-level claims to ordinary users.
- Never merge an Iqama estimate into the prayer-time row without the Estimated label and a confidence range.

- When location accuracy is poor, show the effective coordinate radius and the expected timing sensitivity rather than a generic GPS error.
- Use the same information hierarchy on iOS and Android, while using native permission language and platform-specific remediation steps.
- Keep the home experience quiet: no feed, advertisements, unrelated engagement prompts or mandatory sign-in.

4 The accuracy contract

The application must distinguish mathematical precision from real-world agreement. A solar-position algorithm can be extremely precise while a published timetable differs because of twilight criteria, elevation assumptions, refraction, rounding, safety offsets or an authority convention. The product therefore commits to traceability and measured agreement, not a universal claim of second-perfect religious time.

Layer	Question answered	Examples of uncertainty
Astronomical	Where is the Sun relative to the observer and horizon	Ephemeris approximation, refraction model, pressure, temperature
Jurisprudential rule	Which solar event defines the prayer boundary	Fajr and Isha angle, Asr shadow factor, polar handling
Authority convention	How the jurisdiction publishes the time	Offsets, fixed intervals, rounding, city reference point
Location	Which observer coordinates and elevation apply	Horizontal accuracy, GNSS altitude, DEM resolution
Civil time	Which local clock representation applies	Time-zone polygon, law change, DST transition
Device delivery	Whether an alert is presented at the chosen time	Permission, battery state, reboot, silent mode, OEM policy

Release decisions use an error budget. Each discrepancy is assigned to one layer, reproduced from immutable inputs and either explained by an approved rule or treated as a defect. A hidden manual offset is never an acceptable fix.

5 Exact location Geo Salah Engine

The engine calculates each observer independently. It does not take a city time and add a distance-based correction. Home, selected mosque and travel location are separate calculation contexts that share the same date and policy pipeline.

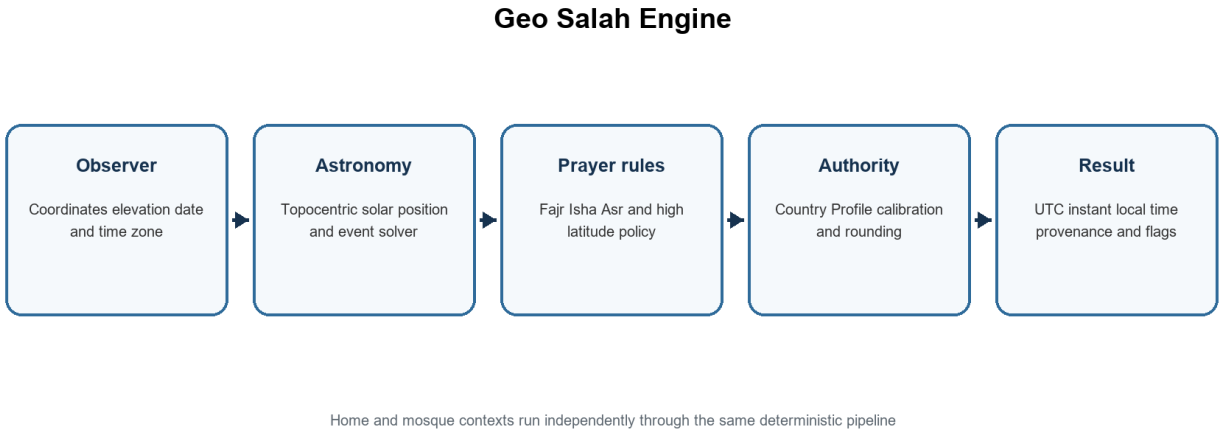


Figure 1 Geo Salah calculation pipeline

Input group	Required fields	Validation
Observer	Latitude, longitude, coordinate source, horizontal accuracy, timestamp	Bounds, freshness, accuracy class and consent state
Terrain	Elevation, vertical datum, source and resolution	Plausibility, coverage and source precedence
Civil time	IANA zone identifier and TZDB version	Polygon match and transition lookup for the event instant
Date	Local civil date and calendar conversion metadata	Unambiguous local day, including DST transitions
Policy	Country Profile, authority variant, school and user overrides	Signed schema, compatibility and effective dates
Environment	Optional pressure and temperature	Used only when trustworthy; otherwise profile defaults

5.1 Calculation contexts

Context	Coordinates	Purpose	Default presentation
My location	Current or saved location	Personal prayer entry and alerts	Recommended authority-aligned result
Selected mosque	Mosque centroid or verified entrance coordinate	Prayer entry at mosque location	Calculated entry plus separate mosque or Iqama data
Manual place	User-selected map point or locality centroid	Privacy-preserving or pre-travel planning	Accuracy reflects coordinate resolution
Benchmark	Fixed lab coordinate and elevation	Regression and authority comparison	Full unrounded trace

6 Astronomical algorithm architecture

6.1 Business explanation

The astronomical core determines solar position for an exact observer and instant. A prayer-rule layer then finds the time at which the relevant event occurs, such as solar transit, a twilight altitude or the Asr shadow condition. A policy layer applies the jurisdiction's approved convention. Keeping these layers separate lets the team improve mathematics without changing policy and update policy without rewriting astronomy.

6.2 Technical pipeline

1. Convert the requested local civil date to a search interval in UTC using the resolved IANA time zone.
2. Compute solar geocentric and topocentric quantities with an internally controlled implementation based on a published high-precision reference such as NREL SPA.
3. Apply observer longitude, latitude, elevation and the approved atmospheric-refraction model to derive apparent solar altitude, azimuth and transit.
4. Solve event roots with a bracketed numerical method. Fajr and Isha solve for configured twilight altitude; sunrise and sunset solve for the approved apparent horizon; Dhuhr uses solar transit; Asr solves the configured shadow ratio.
5. Apply high-latitude fallback only when the primary event has no valid root or violates the profile's boundary conditions.
6. Apply authority adjustments, rounding and display precision in a deterministic order defined by the Country Profile.
7. Emit the final instant together with a machine-readable trace, confidence flags and versions for every input package.

6.3 Calculation order and invariants

Stage	Operation	Invariant
Ephemeris	Solar coordinates for UTC instant	Independent of prayer convention
Topocentric	Observer correction and apparent position	Coordinate and elevation sources recorded
Event solver	Root or transit search	Converges within configured time tolerance or returns an explicit no-event state
Religious rule	Fajr, Isha and Asr criterion	Rule identifier and parameters are explicit
Exceptional rule	High-latitude alternative	Never used silently when a primary event exists
Authority policy	Offsets and fixed intervals	Adjustment reason and effective date are traceable
Rounding	Published minute representation	Applied last and never reused as an astronomical input

6.4 Precision and error budget

Source	Typical treatment	Control
Numerical solver	Sub-second internal tolerance	Golden vectors and convergence assertions
Solar model	Published algorithm bounds	Independent implementation comparison
Refraction	Largest near the horizon and weather-sensitive	Profile default plus documented limitations

Source	Typical treatment	Control
Elevation	Affects apparent horizon, especially sunrise and sunset	GNSS quality gate and DEM fallback
Coordinates	Timing sensitivity varies by event and latitude	Record horizontal accuracy and run perturbation tests
Authority convention	Often dominates user-visible differences	Versioned profile and benchmark evidence
Rounding	Up to approximately one minute depending on convention	Explicit mode such as nearest, floor or ceiling

The application stores unrounded UTC instants and derives local display values at the edge. This prevents cumulative errors when alarms, widgets or travel recalculations reuse prayer times.

7 Country Profiles

A Country Profile is a signed, versioned policy package that translates the astronomical result into the convention expected in a jurisdiction. It may contain national defaults and smaller authority variants when one country uses more than one official method. The profile is data, not executable code.

Profile area	Examples
Identity	Profile ID, jurisdiction, authority, version, effective dates, status
Core criteria	Fajr angle, Isha angle or interval, sunrise and sunset definition, Asr school
Adjustments	Per-prayer offset, safety margin, Ramadan rule, fixed interval
Civil-time policy	IANA zones, border exceptions, official city reference convention
High-latitude policy	Trigger and fallback order with maximum bounds
Rounding	Nearest minute, floor, ceiling or authority-specific publication rule
Evidence	Source documents, benchmark locations, test period and reviewer approval
Compatibility	Minimum engine schema and mobile version

7.1 Profile lifecycle

1. Draft from authoritative public material and record every interpretive decision.
2. Run historical, seasonal and geographic benchmarks against the authority evidence set.
3. Review calculation, jurisprudential interpretation, licensing and UX consequences.
4. Sign and publish to the staging channel, then execute mobile compatibility tests.
5. Promote gradually to production. Retain the previous package for instant rollback.
6. Retire only after all supported clients have moved past its effective period and evidence retention obligations are met.

8 Authority Calibration Engine

Calibration is an offline engineering process, not an adaptive offset applied on a user's phone. The lab compares engine output with approved authority data, explains the difference and proposes a profile change only when the convention is understood and reproducible.

Step	Output	Release rule
Acquire	Immutable source snapshot with license and retrieval metadata	Source must be legally usable and traceable
Normalize	Canonical date, location, prayer and published minute	No inferred values without a recorded transform
Calculate	Unrounded engine trace for the same context	Pinned engine, profile, TZDB and elevation versions
Compare	Signed discrepancy in seconds and minutes	Compare before and after publication rounding
Classify	Astronomy, policy, location, civil time, source or unknown	Unknown discrepancies block promotion above threshold
Approve	Calibration report and profile change	Two-person review for production profile
Monitor	Trend by prayer, city, season and version	Automatic rollback when a canary threshold is breached

8.1 Prohibited calibration practices

- Adding a global minute because several examples appear late without explaining the governing convention.
- Training directly on third-party app output or scraping a competing application as the runtime truth source.
- Allowing a machine-learning model to alter prayer boundaries without a deterministic, reviewable rule.
- Combining multiple authorities into an average and calling the result verified.
- Changing historical calculations after a profile update without preserving the version originally shown to the user.

9 GPS and elevation resolver

The resolver chooses the best available observer state and reports its quality. Horizontal position and elevation have separate confidence because mobile GNSS altitude is often less stable than latitude and longitude.

Priority	Source	Use rule	Metadata retained
1	Fresh device location	Use horizontal coordinates when accuracy and age meet the active context threshold	Provider, age and horizontal accuracy
2	Quality device altitude	Use only when vertical accuracy is reported and passes the profile threshold	Ellipsoid or orthometric datum and vertical accuracy
3	Offline DEM	Sample locally cached SRTM-derived tile or equivalent open DEM	Dataset version, tile, cell size and interpolation
4	Manual location	Use user-selected coordinate; label its resolution	Map point or locality centroid
5	Last known location	Temporary fallback within policy age limit	Age and stale warning

9.1 Location change policy

- Recalculate when movement could change the displayed minute, the time zone changes, or a user explicitly refreshes.
- Do not continuously track in the background merely to improve an already stable local schedule.
- Use geofencing sparingly and only for user-enabled Travel Mode; store geofences locally.
- When precise location is denied, offer manual place selection and explain the expected loss of precision.

10 Time zone and daylight saving time

Coordinates resolve to an IANA time-zone identifier through an offline polygon dataset. The event instant is then converted using a pinned IANA TZDB release. A fixed numeric UTC offset is never stored as the governing rule.

Case	Required behavior
Ordinary day	Resolve zone and apply the offset valid at each prayer instant
DST transition	Calculate in UTC, then convert each event; never add 24 hours to yesterday's local value
Government change	Publish a signed TZDB package and priority update; retain previous version for audit
Border ambiguity	Use polygon result, show zone in Accuracy Passport and permit user correction
Offline after law change	Show the last package date and a targeted warning when the jurisdiction is affected
Travel	Cancel or supersede old-zone alarms and schedule the new context after confirmation policy

11 High latitude and missing event rules

At high latitudes, one or more twilight events may not occur on a civil date. The engine returns a typed no-event state before applying any alternative. The Country Profile selects the approved fallback and the interface identifies it as an alternative calculation.

Method	Definition	Use
Middle of night	Half of the interval from sunset to sunrise	Simple fallback where accepted by the governing profile
One seventh	One seventh of the night allocated near each boundary	Profile-specific alternative
Angle portion	Twilight angle divided by 60 as a proportion of night	Common configurable method
Nearest valid day	Use a nearby date on which the event exists	Only with bounded search and explicit authority approval
Nearest latitude	Calculate at an approved lower latitude while preserving longitude	Exceptional profile rule with clear disclosure
Authority timetable	Use a verified published value	When the jurisdiction provides an approved alternative

The evaluation corpus must include polar day, polar night, event disappearance, event reappearance and dates near the threshold. A fallback must never create a prayer ordering violation.

12 Selected mosque calculation

Selecting a mosque creates a second observer context using that mosque's coordinates and the same policy version. The result is the calculated prayer entry at the mosque location. It is not evidence of the actual loudspeaker time or the mosque's Iqama schedule.

Displayed value	Source	Confidence label
Prayer entry at my location	Geo Salah at user coordinate	Calculated or Authority aligned
Prayer entry at selected mosque	Geo Salah at mosque coordinate	Calculated
Mosque published Adhan	Approved current mosque or authority source	Verified
Mosque clock offset	Observed and curated difference from calculated entry	Observed, never Verified unless approved
Iqama	Approved source or estimate model	Verified or Estimated

12.1 Mosque coordinate sources

- Open geographic datasets may seed candidate mosques, subject to attribution and license review.
- Coordinates receive source, precision, last-reviewed date and duplicate-resolution metadata.
- User corrections enter a moderation queue and do not overwrite the production record directly.
- The product remains functional without a mosque directory. Mosque discovery is an enhancement, not a calculation dependency.

13 Iqama model and verified data separation

The initial product estimates Iqama as a range after the prayer entry, normally using a profile default such as 10 to 15 minutes. This is intentionally separate from a mosque's actual congregational time. The UI must make the distinction visible at the point of use.

Class	Definition	Example	Expiry
Verified Iqama	Current data from an approved, traceable source	Mosque-published Dhuhr at 12:30	Source-specific, usually daily or seasonal
Estimated range	Rule-based window after calculated or published Adhan	Expected 12:25 to 12:30	Recomputed each day
Community observation	User-submitted report awaiting validation	Observed around 12:28	Short-lived and never promoted silently
Unknown	No defensible value	No Iqama shown	Until a rule or verified source exists

13.1 Future estimation model

A later model may estimate by country, locality, prayer, weekday, season and Ramadan state. It must produce a range and confidence score, retain the contributing observations, resist abuse and pass a bias review. It may improve usefulness but cannot change prayer entry times or use the word Verified.

14 Qibla

Qibla direction is calculated locally as the initial great-circle bearing from the observer to the Kaaba reference coordinate. The numeric bearing is distinct from the phone's compass heading. The experience should therefore show both the calculated bearing and sensor quality.

Component	Design
Geodesic calculation	Double-precision bearing with normalized longitude difference and golden test vectors
Reference coordinate	Versioned Kaaba coordinate stored in the engine configuration
Device orientation	Fused magnetometer, accelerometer and gyroscope where available
Calibration	Platform-native guidance for figure-eight movement or sensor reset
Interference	Warn when magnetic field or heading accuracy is poor
Fallback	Show map bearing or numeric degrees when a reliable compass is unavailable
Privacy	No location upload is required

15 Athan reliability architecture

Alarm reliability is implemented with native platform services because the operating system controls delivery. Flutter owns settings, shared business rules and the common experience. Swift and Kotlin adapters own scheduling, permission checks, reboot recovery, system surfaces and delivery receipts that each platform exposes.

15.1 Common scheduler contract

Operation	Contract
schedule	Accept immutable prayer instant, audio asset, behavior policy and idempotency key
cancel	Cancel by schedule generation and prayer identity
reconcile	Compare desired future schedule with platform state and repair differences
health	Return permissions, audio, battery, notification and exact-scheduling capabilities
test	Schedule a short user-initiated test and record the observed result
events	Report scheduled, presented, user-dismissed or unknown states without claiming unavailable OS evidence

15.2 Android design

- Use AlarmManager exact scheduling only for user-enabled, time-critical Athan. Select the permitted API after legal and store-policy review; check exact-alarm capability at runtime.
- Use PendingIntent-based delivery so the alert survives the application process. Reschedule after boot, time-zone change, time change, package replace and permission-state change.

- Use a notification channel configured for the selected Athan behavior. Validate audio availability and channel settings before claiming Ready.
- Provide OEM-specific battery guidance as maintainable help content, not as hard-coded promises. Fall back visibly when exact scheduling is unavailable.
- Generate only the necessary scheduling horizon, for example seven days, then reconcile daily to reduce stale alarms after travel or profile updates.

15.3 iOS design

- Use AlarmKit on supported iOS versions after user authorization because it provides prominent scheduled alarms and can override Focus and silent mode when the platform permits.
- Maintain a capability adapter for older supported iOS versions using local notifications, with a clearly lower assurance state.
- Schedule a bounded future horizon and refresh while the app is active or through permitted background opportunities. Do not depend on unrestricted background execution.
- Use App Groups for widgets and watch extensions to read the latest signed local schedule without network access.
- Athan Test is user initiated, time bounded and reports what the OS actually presented. It never records microphone evidence or private ambient audio.

15.4 Athan Health state model

State	Definition	UX action
Ready	Required permissions, exact capability, audio and future schedule are confirmed	Show next alert and last reconciliation time
Limited	Alert can be scheduled but one condition may reduce prominence or timing	Explain the exact limitation and remediation
Not ready	No valid future alert can be scheduled	Guide permission or settings repair
Unknown	The platform does not expose enough evidence	Do not imply readiness; offer Athan Test

16 Offline first design

The mobile application can calculate, display and schedule without a network connection. Connectivity improves freshness and optional discovery but is not required for the daily prayer experience.

Local asset	Purpose	Update model
Astronomical engine	Deterministic calculation	Application release with conformance version
Country Profiles	Jurisdiction rules and calibration metadata	Signed small packages with rollback
TZDB and polygons	Time-zone resolution and offsets	Bundled baseline plus signed urgent updates
Elevation tiles	Terrain fallback	Regional on-demand cache with packaged core coverage
Prayer schedule	Fast UI, widgets and alarms	Rolling cache recalculated after input changes
Athan audio	Offline playback	Bundled default plus user-downloaded licensed assets

Local asset	Purpose	Update model
Mosque index	Optional nearby discovery	Regional compressed packages with source attribution

16.1 Conflict and rollback rules

- A newly downloaded package is verified, staged and activated atomically. Partial downloads never replace an active package.
- The scheduler records the package generation used. If an update changes future times, it reconciles only future alarms and preserves the audit record for prior ones.
- A remote configuration service may choose a signed package version but cannot send arbitrary prayer times or executable rules.
- The last known good package remains available for rollback until the next version is proven stable.

17 Extended experience

Capability	User value	Architecture note	Release
Home and timeline	Immediate next-prayer context	Shared Flutter UI and local schedule	MVP
Accuracy Passport	Explains the displayed time	Calculation trace rendered for people	MVP
Athan Health and Test	Confidence before relying on an alert	Native platform adapters	MVP
Qibla	Direction with sensor-quality guidance	Local geodesic plus native sensors	MVP
Home and lock widgets	Prayer time without opening the app	Shared local snapshot via native extensions	V1
Apple Watch and Wear OS	Glanceable schedule and next prayer	Companion sync, local fallback	V1.5
Travel Mode	Safe schedule changes across place and time zone	Context versioning and alarm reconciliation	V1
Ramadan Mode	Suhoor, fasting window and Ramadan-specific Isha rules	Profile-driven seasonal policy	V1
Advanced prayer windows	Makruh, Duha, midnight and last third where defined	Separate rule module with scholarly and profile review	V1.5
Selected mosque	Location-specific entry plus distinct Iqama information	Optional open-data mosque index	V1.5
Family and home coordination	Avoid duplicate Athan across opted-in devices	Privacy-preserving local or user-controlled pairing	V2

18 Privacy and security

18.1 Privacy model

- No login is required for calculation, Qibla, local alarms, widgets or downloaded data packages.
- Precise location is processed on device and stored only when the user saves a place or enables a location-dependent feature.
- Analytics use coarse event counts and never include raw latitude, longitude, advertising identifiers, contact lists or a persistent cross-app identity.

- Diagnostic upload is opt in, redacts coordinates by default and shows the exact package before submission.
- Settings export is local and user initiated. Cloud backup, if introduced later, remains optional and encrypted.

18.2 Threats and controls

Threat	Control	Evidence
Tampered profile or TZDB package	Ed25519 signature, hash, schema validation and monotonic version policy	Package verification test and key-rotation drill
Compromised update service	Offline signing, least privilege, staged channels and client rollback	Audit log and canary report
Malicious deep link or intent	Strict allowlist, non-exported Android components and validated routes	Mobile security tests
Alarm spoofing	Unpredictable identifiers, signed local schedule generation and restricted receivers	Platform instrumentation tests
Sensitive local data exposure	Platform-protected storage for settings and least-data schema	Storage review and device backup test
Supply-chain compromise	Locked dependencies, SBOM, license scan, provenance and reproducible checks	CI attestation
Community data abuse	Rate limits, moderation, reputation and separation from verified records	Abuse test and moderation audit

19 Logical architecture

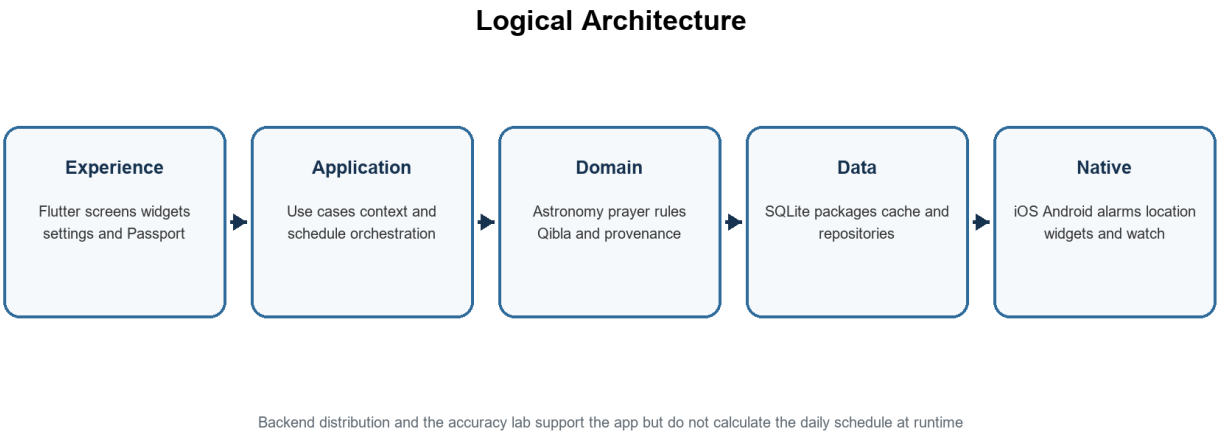


Figure 2 Logical architecture

Layer	Components	Ownership rule
Experience	Home, timeline, Accuracy Passport, Qibla, mosque, settings and health	Flutter presentation; platform surfaces remain native
Application	Use cases, context selection, schedule orchestration and feature policy	No UI or storage dependencies
Domain	Astronomy, prayer rules, calibration application, Qibla and provenance	Pure deterministic code with conformance tests
Data	Repositories, SQLite, package store, cache and update client	Atomic versioned reads and migrations

Layer	Components	Ownership rule
Native platform	AlarmKit, AlarmManager, notifications, location, compass, widgets and watches	Typed platform contracts and capability tests
Backend	Package registry, distribution, evidence catalog, optional mosque ingestion and aggregate telemetry	No dependency for core calculation
Accuracy lab	Reference ingestion, benchmark runner, discrepancy triage and reports	Separate controlled environment

19.1 Recommended technology baseline

Area	Recommendation	Reason
Mobile UI and shared logic	Flutter and Dart	Consistent cross-platform design with one product codebase
iOS integration	Swift	Direct access to AlarmKit, WidgetKit, Live Activities, Core Location and watchOS
Android integration	Kotlin	Direct access to AlarmManager, notifications, services, widgets and Wear OS
Local database	SQLite	Mature transactional storage with offline query support
Backend	Small stateless service in an approved open-source stack	Package metadata, evidence and optional data ingestion only
Object distribution	Static object storage plus CDN or equivalent	Low cost, cacheable signed packages
Lab	Python or JVM tooling plus independent calculation implementations	Strong numerical and data-analysis ecosystem

20 Deployment architecture

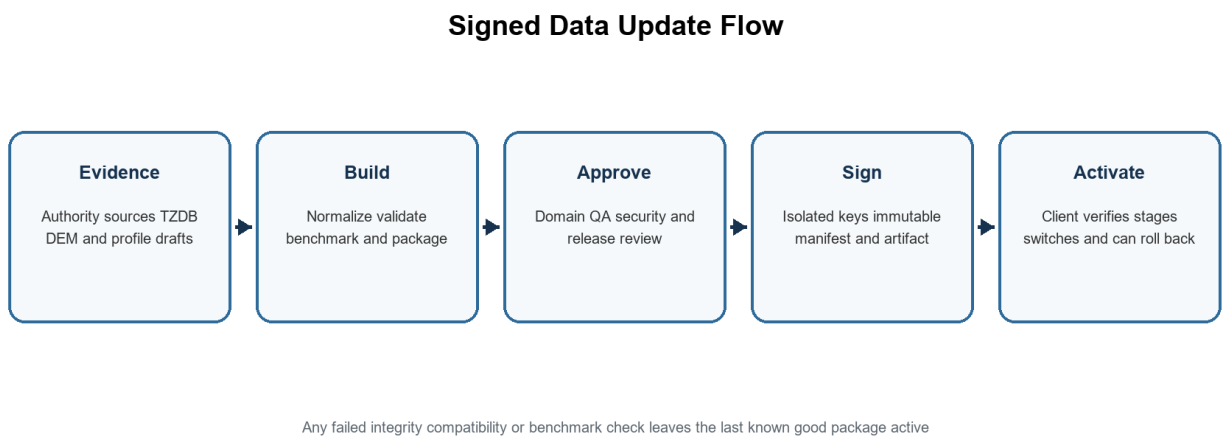


Figure 3 Deployment and signed update flow

Production separates authored policy from distributed packages. Reviewers approve evidence and profile content. A controlled build produces deterministic artifacts. An offline signing step signs release manifests. Clients verify signatures before activation. The mobile application continues using the last known good package if any network, integrity or compatibility check fails.

Zone	Contents	Trust boundary
Mobile device	Engine, local database, alarms, widgets, audio and cached packages	User-controlled device; verify all downloaded content
Public distribution	Immutable signed packages and manifests	Assume transport or cache may be hostile
Control plane	Profile authoring, evidence catalog, release metadata and approvals	Authenticated staff access and audit
Signing	Offline or isolated signing keys	Highest assurance; dual control and rotation
Accuracy lab	Reference datasets, benchmark runners and reports	No production write access except approved artifacts
Telemetry	Coarse aggregates and opt-in diagnostics	Separated from package publication

21 Data model

Entity	Key fields	Notes
LocationContext	id, lat, lon, horizontalAccuracy, elevationRef, zoneId, source, observedAt	Home, current, manual, mosque or benchmark
ElevationSample	value, datum, source, resolution, verticalAccuracy, datasetVersion	Separates device and DEM evidence
CountryProfile	id, jurisdiction, authority, version, effectiveRange, ruleSet, signature	Immutable after publication
PrayerRuleSet	fajr, isha, sunrise, sunset, dhuhr, asr, highLatitude, rounding	Schema validated and human reviewable
CalculationResult	contextId, date, prayer, utcInstant, localTime, provenanceId, flags	Stores unrounded instant
CalculationProvenance	engine, profile, tzdb, elevation, inputsHash, adjustmentTrace	Accuracy Passport source
Mosque	id, name, coordinate, source, precision, reviewedAt	Optional open or curated directory
IqamaRecord	mosqueId, prayer, valueOrRange, class, source, confidence, effectiveRange	Class is Verified, Estimated, Observed or Unknown
AlarmPlan	generation, prayerResultId, platformId, state, scheduledAt, reconciledAt	Idempotent scheduling and audit
DataPackage	type, version, schema, effectiveRange, hash, signature, activatedAt	Atomic activation and rollback
Discrepancy	benchmarkId, prayer, expected, actual, delta, classification, owner, status	Accuracy lab workflow

21.1 Data integrity rules

- CalculationResult references immutable provenance; it is never updated in place after publication to another component.
- Verified and Estimated Iqama records use different types and cannot be converted by a display-layer flag.

- Package version and schema compatibility are checked before a transaction changes the active pointer.
- Local retention removes obsolete schedules and diagnostics while preserving only the minimum evidence needed to explain recent alarms.

22 APIs and contracts

Core calculation uses in-process domain APIs. Network APIs distribute signed content and accept optional evidence or aggregate telemetry. The mobile client does not call a prayer-time web API for normal operation.

Interface	Method or event	Purpose	Offline behavior
Engine	calculateDay context profile date	Return ordered results and provenance	Fully available
Engine	calculateQibla context	Return bearing and reference version	Fully available
Scheduler	reconcile desiredPlan	Make platform alarms match desired schedule	Fully available
Package API	GET manifest channel	Discover compatible signed versions	Use cached manifest
Package API	GET package type version	Download immutable artifact	Use active package
Evidence API	GET profile evidence	Support audit and profile review	Optional cached summary
Mosque API	GET regional package	Download optional mosque index	Use cached region or omit
Feedback API	POST correction	Submit explicit user correction	Queue locally with consent
Telemetry API	POST aggregate batch	Send privacy-filtered operational counts	Queue or discard by retention policy

22.1 API standards

- Use HTTPS, explicit API versions, short timeouts, idempotency for submissions and RFC-style structured errors.
- Immutable package URLs include content hashes and support cache validation.
- Manifests declare minimum client, schema, size, hash, signature, effective date, emergency priority and rollback target.
- Network failure never deletes an active package or blocks today's calculation.
- Diagnostics and corrections require explicit consent and return a local receipt ID without creating an account.

23 Dependency and dataset policy

Production must not depend at runtime on a paid prayer-time API, a competing prayer application, a mandatory mosque partnership or an advertising network. Open-source software and public or openly licensed datasets may be used after security, maintenance and license review. Critical domain behavior remains under internal control.

Category	Preferred approach	Release gate
Astronomy	Internal deterministic implementation validated against published references	Independent comparison and golden corpus
Time zones	IANA TZDB plus an openly licensed coordinate-to-zone dataset	Version pin, license review and transition tests

Category	Preferred approach	Release gate
Elevation	NASA SRTM-derived or equivalent open DEM packages	Coverage, datum and redistribution review
Mosques	Optional OpenStreetMap-derived regional extracts or public authority data	ODbL attribution and produced-database review
Mobile framework	Flutter stable channel with locked SDK and package versions	Compatibility, security and license scan
Storage	SQLite and thin audited bindings	Migration and corruption recovery tests
Audio	Internally licensed or permissively licensed files	Rights, attribution and offline playback test
Maps	Optional open map rendering or static local representation	No core dependency on a paid map SDK

23.1 Software bill of materials

Every release produces an SBOM, notices file, vulnerability report and dependency-diff review. Packages with unclear maintenance, network behavior or restrictive licenses are rejected or isolated behind replaceable adapters. A dependency cannot determine prayer rules without conformance tests owned by the project.

24 Update strategy

Asset	Cadence	Urgent trigger	Distribution
Country Profiles	Quarterly review and as authorities change	Published rule, offset or Ramadan change	Signed data package
TZDB	Monitor every upstream release	Government time-law change affecting supported users	Priority signed package plus later app baseline
Time-zone polygons	At least quarterly	Boundary correction or new zone mapping	Signed package
Elevation	Annual or source release	Material tile correction	Regional package
Astronomical engine	Normal app release	Validated correctness or security defect	Application binary; emergency patch if required
Mosque data	Regional schedule based on source	Duplicate, coordinate or closure correction	Optional regional package
Athan audio	Curated releases	Rights withdrawal or corrupted asset	Signed asset manifest

24.1 Safe rollout

1. Detect an upstream change and open a tracked update record with affected countries, dates and client versions.
2. Build the candidate package reproducibly and run schema, signature, regression and benchmark suites.
3. Release to internal, accuracy-lab and beta channels before a small production canary.
4. Compare discrepancies, crash-free activation, alarm reconciliation and package uptake.
5. Promote globally or by affected region. Retain a remote kill switch that selects the last known good signed package.
6. Publish a human-readable change note in Accuracy Passport when displayed times may change.

25 Accuracy and evaluation lab

25.1 Benchmark design

The lab combines fixed astronomical vectors, authority comparisons and end-to-end device scenarios. It covers seasons, latitudes, elevations, time-zone transitions and known policy differences. Each benchmark stores the expected source and the calculation inputs so a result can be reproduced years later.

Region	Example benchmark cities	Why included
Middle East	Dubai, Abu Dhabi, Makkah, Riyadh, Muscat	Core target conventions, arid refraction assumptions and authority profiles
North Africa	Cairo, Alexandria, Aswan, Casablanca, Tunis	Longitude spread, coastal and inland cases, national authority comparisons
Europe	London, Paris, Oslo, Stockholm, Reykjavik	DST and progressive high-latitude behavior
Asia	Karachi, Delhi, Kuala Lumpur, Jakarta, Tokyo	Multiple schools, humid tropics and broad longitude coverage
North America	New York, Toronto, Chicago, Vancouver, Anchorage	Multiple zones, DST and high latitude
Southern hemisphere	Cape Town, Johannesburg, Sydney, Auckland, Ushuaia	Reversed seasons and southern high latitude
Extreme tests	Tromso, Longyearbyen, Utqiagvik, equator and date-line points	Missing events, reappearance, polar conditions and zone boundaries

25.2 Dataset dimensions

- At least the solstices, equinoxes and one mid-month date per season for every benchmark location.
- Every DST transition and at least seven days before and after it in affected zones.
- Elevation perturbations at sea level, local DEM value and a high-observer scenario.
- Coordinate perturbations matching 10 m, 100 m, 1 km and locality-centroid uncertainty.
- All supported Fajr, Isha, Asr, rounding and high-latitude profile variants.
- Historical regression cases for every production discrepancy and incident.

25.3 Discrepancy report

Field	Example content
Identity	Case ID, city, coordinate, elevation, date, prayer
Versions	Engine, Country Profile, TZDB, polygon, DEM and authority dataset
Values	Expected published minute, calculated raw instant, displayed minute and signed delta
Trace	Solar event, rule parameters, fallback, offsets and rounding
Classification	Astronomy, policy, civil time, location, source quality, platform or unknown
Decision	Accepted convention, defect, data correction or investigation
Ownership	Named role, due date, reviewer and linked change

25.4 Release thresholds

- Zero prayer ordering failures, NaN results, unhandled missing events or time-zone conversion ambiguities.
- All golden astronomical vectors within the approved numerical tolerance.
- All Country Profile benchmarks meet their declared tolerance or carry an approved explained exception.
- No unexplained discrepancy over two minutes in launch countries; tighter profile-specific thresholds may apply.
- Identical conformance results on Dart, lab reference and native boundary tests for the same input package.
- Device alarm suite passes supported OS versions and a representative OEM matrix before store submission.

26 Observability and auditability

Observability must help diagnose calculation and delivery without creating a location-tracking system. Most evidence stays on device and is converted into privacy-safe aggregates only when needed.

Signal	Location	Retention	Privacy rule
Calculation trace	Device	Recent schedules and user-visible history	Contains precise location; no automatic upload
Package activation	Device and aggregate service	Version lifecycle	No coordinate
Alarm plan state	Device	Short operational window	No audio or ambient recording
Athan Test result	Device; optional aggregate	User-controlled diagnostic period	No persistent identity
Discrepancy report	Accuracy lab	Long-term release evidence	Fixed benchmark locations only
Security audit	Control plane	Policy-defined	Access controlled and tamper evident

26.1 Audit identifiers

Every displayed prayer time can be reconstructed from a compact Calculation ID that resolves locally to input hash, engine version, Country Profile, time-zone package, elevation source and adjustment trace. Support staff can ask the user to export a redacted diagnostic bundle without requiring an account.

27 Nonfunctional requirements and testing

27.1 Nonfunctional requirements

Quality	Requirement
Availability	Core calculation, schedule, Qibla and alarms work offline after installation
Performance	Cold display of cached home under 1 second on target devices; one-year schedule calculation under 500 ms in benchmark profile
Accuracy	Meets Section 25 release thresholds with deterministic results
Reliability	Alarm plan reconciles after reboot, time change, travel, profile update and permission change
Battery	No continuous background GPS; background work is event driven and measured

Quality	Requirement
Security	Signed packages, locked dependencies, encrypted transport and platform-protected local storage
Privacy	No account, advertising ID or automatic precise-location upload for core operation
Accessibility	WCAG-aligned contrast, scalable text, screen-reader labels, motion reduction and non-color status cues
Localization	Arabic and English from MVP; RTL tested; locale does not alter calculation semantics
Maintainability	Domain modules have at least 90 percent branch coverage for critical rules and documented ADRs
Compatibility	Support matrix defined per release; degraded capability is explicit on older OS versions
Recoverability	Rollback to last known good data package without reinstalling the application

27.2 Test strategy

Level	Coverage
Unit	Solar helpers, root solver, prayer rules, high-latitude methods, rounding, Qibla and schema validation
Property based	Ordering, range, continuity, longitude shifts, date boundaries and no-event invariants
Golden conformance	Pinned input and exact trace output across engine versions
Differential	Compare with an independent reference implementation and approved authority sources
Integration	Database migrations, package activation, rollback, location resolver and scheduler reconciliation
Native platform	Permissions, reboot, Doze, Focus, silent mode, audio channel, widgets and watch sync
End to end	First run through displayed prayer, Accuracy Passport, enabled Athan and successful Athan Test
Security	Static analysis, dependency scan, package tampering, intent and deep-link abuse, secret scanning
Accessibility	Screen readers, large text, contrast, RTL, reduced motion and touch targets
Resilience	No network, stale package, corrupted cache, disk pressure, clock change and interrupted update

27.3 Device and OS matrix

- Current and previous two major iOS versions, with an explicit capability split for AlarmKit support.
- Current Android version plus supported older versions, including Pixel, Samsung and at least two high-volume OEM battery-management variants.
- Low-memory and small-screen devices, tablets where supported, Arabic RTL and large accessibility text.

- Physical devices are mandatory for alarm, compass, reboot, battery and wearable tests; simulators are insufficient.

28 CI CD and release controls

Pipeline stage	Automated gates	Manual approval
Pull request	Format, lint, unit, property, schema, secret and license checks	Code review and domain review for rule changes
Merge	Golden corpus, differential tests, mobile integration and SBOM	Architecture decision when contract changes
Nightly	Full benchmark cities, seasonal matrix and device smoke tests	Accuracy triage for threshold breach
Release candidate	Signed-package verification, migration, rollback, accessibility and performance	Product, QA, accuracy, security and release owner
Beta	Crash-free rate, activation, alarm health and opt-in diagnostics	Go or no-go review
Production canary	Regional uptake, discrepancy and scheduler health	Promotion or rollback
Store release	Binary provenance, notices, privacy labels and signing verification	Authorized release manager

28.1 Branching and environments

- Use trunk-based development with short-lived branches and protected main.
- Separate development, lab, beta and production package channels. A mobile build can subscribe only to channels permitted by its signing identity.
- Keep signing secrets outside source control, use short-lived CI credentials and require dual control for production data-package signing.
- Store benchmark evidence and generated reports as immutable release artifacts.

29 Release roadmap

Release	Outcome	Included capabilities	Exit gate
MVP	Prove trusted local prayer times and dependable basic Athan	Exact-location engine, launch Country Profiles, TZDB, DEM fallback, home, Accuracy Passport, Qibla, iOS and Android Athan Health and Test, offline cache, Arabic and English	Accuracy thresholds met in launch markets and controlled device alarm pass
V1	Complete the everyday prayer experience	Widgets, lock surfaces, Travel Mode, Ramadan Mode, advanced update operations, broader profiles, accessibility hardening and public beta learning	Stable data update process and target reliability metrics
V1.5	Add place and wearable context	Selected mosque calculation, clearly typed Iqama estimates, mosque regional packages, Apple Watch, Wear OS and advanced prayer windows	Data licensing, moderation and wearable battery gates
V2	Coordinate trusted experiences across chosen devices	Optional verified mosque feeds, privacy-preserving household coordination, richer confidence models and expanded global coverage	Independent privacy and security review plus scale evidence

29.1 Phase plan

Phase	Sprints	Primary decision
0 Foundation	Sprint 0 to 1	Can the team reproduce approved reference results and integrate native alarm spikes
1 Accuracy core	Sprint 2 to 4	Is the engine, profile and civil-time pipeline deterministic and explainable
2 MVP product	Sprint 5 to 6	Can real users calculate, inspect and rely on a tested alert offline
3 V1 experience	Sprint 7 to 10	Can the product serve daily, travel and Ramadan use at release quality
4 V1.5 context	Sprint 11 to 14	Can mosque, Iqama and wearable features preserve provenance and battery goals
5 V2 scale	Sprint 15 to 18	Can global updates and optional multi-device features operate safely at scale

30 Sprint by sprint delivery plan

The plan assumes two-week sprints, a stable cross-functional team and continuous accuracy-lab support. Sprint acceptance is evidence based. A feature is not complete when only the shared Flutter UI works; native platform behavior, offline operation, analytics constraints, accessibility and documentation are part of the definition of done.

Sprint 0 Architecture runway

Scope	Outputs	Acceptance criteria	Dependencies
Confirm launch countries; define ADRs, repositories, coding standards, profile schema and evidence workflow; prototype Dart-to-native contracts.	Approved architecture baseline; CI skeleton; risk spikes; benchmark inventory.	Engine input and output schemas compile; sample signed profile verifies; native channel round trip works on physical iOS and Android.	Product launch-market decision; Apple and Google developer accounts.

Sprint 1 Reference and alarm spikes

Scope	Outputs	Acceptance criteria	Dependencies
Implement independent solar reference harness; test NREL vectors; prototype Android exact alarm and iOS AlarmKit or fallback capability.	Reference harness; first discrepancy report; alarm feasibility matrix.	Selected solar vectors reproduce within tolerance; a user-initiated alarm test presents on supported reference devices; limitations documented.	Sprint 0 schemas; physical device lab.

Sprint 2 Astronomical core

Scope	Outputs	Acceptance criteria	Dependencies
Build solar position, root solver, Dhuhr, sunrise and sunset; add property and golden tests.	Geo Salah core library v0.1; trace format.	Deterministic UTC instants; no ordering or convergence failures across baseline corpus.	Reference harness.

Sprint 3 Prayer rules and profiles

Scope	Outputs	Acceptance criteria	Dependencies
Add Fajr, Isha, Asr, rounding and Country Profile interpreter; author first launch profiles.	Rule engine; signed profile candidates; review checklist.	All rule variants pass conformance tests; no executable expressions in profiles; profile evidence complete.	Astronomical core; authority source material.

Sprint 4 Location civil time and elevation

Scope	Outputs	Acceptance criteria	Dependencies
Implement location quality model, zone polygon resolution, TZDB conversion, DEM sampling, DST and high-latitude methods.	Observer resolver; time-zone and elevation packages; exceptional-event suite.	Correct transition-day conversion; DEM fallback works offline; every missing event produces an explicit state and approved fallback.	Profile schema; licensed datasets.

Sprint 5 MVP experience and Qibla

Scope	Outputs	Acceptance criteria	Dependencies
Build onboarding, home, daily timeline, manual location, Accuracy Passport and Qibla with sensor quality.	Arabic and English core UX; local database; accessible designs.	First run reaches a calculated schedule offline; Passport reconstructs result; Qibla golden vectors pass; RTL and screen reader review passes.	Engine APIs; UX content approval.

Sprint 6 MVP Athan and hardening

Scope	Outputs	Acceptance criteria	Dependencies
Complete native schedulers, Athan Health, Athan Test, reboot and time-change recovery, audio, privacy controls and MVP release testing.	MVP beta builds; device evidence; privacy disclosures; release report.	Accuracy and alarm gates pass; no Severity 1 defects; core flows require no account or network; rollback drill succeeds.	Sprints 1 to 5; store permission review.

Sprint 7 Widgets and system surfaces

Scope	Outputs	Acceptance criteria	Dependencies
Add home and lock widgets, Live Activity or Dynamic Island where appropriate, and Android widgets.	Native extensions reading shared local snapshot.	Widgets remain correct offline, across midnight and after profile update; stale state is visible.	MVP schedule store; native extension entitlements.

Sprint 8 Travel Mode

Scope	Outputs	Acceptance criteria	Dependencies
Detect meaningful place or zone change; design confirmation policy; reconcile alarms and saved places.	Travel journeys; context-versioned schedule plan.	No duplicate old-zone alerts; manual override works; privacy and battery tests pass.	Location resolver; scheduler reconciliation.

Sprint 9 Ramadan and update operations

Scope	Outputs	Acceptance criteria	Dependencies
Implement profile-driven Ramadan rules, fasting window, package canary, rollback and user-facing change notes.	Ramadan Mode; production update dashboard; emergency runbook.	Ramadan rule changes are profile tested; package rollback restores prior schedule atomically; change note identifies affected times.	Signed package service; authority review.

Sprint 10 V1 release quality

Scope	Outputs	Acceptance criteria	Dependencies
Expand profiles, tune performance, complete accessibility, localization, device matrix and store readiness.	V1 release candidate; operational dashboards; support playbook.	NFRs met; target benchmark and device matrix green; privacy labels and notices reviewed.	Sprints 7 to 9; beta feedback.

Sprint 11 Mosque data foundation

Scope	Outputs	Acceptance criteria	Dependencies
Define mosque entity, open-data ingestion, deduplication, attribution, regional packages and correction queue.	Mosque pipeline and local search prototype.	License obligations met; duplicates and coordinate precision recorded; application works when mosque package is absent.	Legal license review; package infrastructure.

Sprint 12 Selected mosque experience

Scope	Outputs	Acceptance criteria	Dependencies
Add nearby search, mosque selection and independent coordinate calculation; compare home and mosque entry.	Mosque UX; provenance views; offline regional search.	No UI implies calculated entry equals actual loudspeaker time; calculation context is reproducible; stale mosque data is labeled.	Sprint 11; Geo Salah context API.

Sprint 13 Iqama estimates and observations

Scope	Outputs	Acceptance criteria	Dependencies
Implement 10 to 15 minute default ranges, profile overrides, confidence labels and moderated observations.	Typed Iqama model; estimate UI; moderation workflow.	Verified and Estimated are structurally distinct; estimate always displays range and class; abuse tests pass.	Mosque entity; content policy.

Sprint 14 Wearables and V1.5 release

Scope	Outputs	Acceptance criteria	Dependencies
Deliver Apple Watch and Wear OS next-prayer experience; optimize sync and battery; harden mosque and Iqama features.	V1.5 release candidate; wearable test report.	Watch shows correct cached schedule offline; sync recovers after disconnect; battery and accessibility gates pass.	Platform companion setup; Sprints 11 to 13.

Sprint 15 Advanced prayer windows

Scope	Outputs	Acceptance criteria	Dependencies
Add Duha, Islamic midnight, last third and caution windows with reviewed definitions and separate display.	Advanced rule module and educational UX.	Definitions are profile controlled, never alter obligatory prayer entry, and pass high-latitude tests.	Domain review; engine extension points.

Sprint 16 Optional verified sources

Scope	Outputs	Acceptance criteria	Dependencies
Create approval, freshness and revocation workflow for public authority or mosque-published schedules without mandatory partnerships.	Verified-source adapter; provenance and expiry controls.	Expired or revoked data cannot display as Verified; source outage leaves calculated schedule intact.	Data governance; source agreements where available.

Sprint 17 Private multi-device coordination

Scope	Outputs	Acceptance criteria	Dependencies
Prototype optional household pairing and duplicate-Athan suppression with local-network or end-to-end encrypted control.	Threat model; pairing UX; opt-in coordinator.	Core app remains accountless; leaving a group removes keys; no third party can infer household location from coordination metadata.	Security review; device protocol decision.

Sprint 18 V2 scale and independent review

Scope	Outputs	Acceptance criteria	Dependencies
Expand global profiles, load test distribution, run external security and privacy assessment, perform disaster recovery and final accuracy audit.	V2 candidate; independent findings; global operations handbook.	All critical findings closed; regional rollback proven; benchmark thresholds met; support and incident owners accept handover.	All prior phases; external review availability.

31 Delivery governance and RACI

Deliverable	Business	Product	Accuracy lead	Architect	Mobile	Backend data	QA	Security
Vision and release scope	A	R	C	C	C	C	C	C

Deliverable	Business	Product	Accuracy lead	Architect	Mobile	Backend data	QA	Security
Country Profile	I	C	A R	C	I	R	C	C
Astronomical engine	I	C	A	R	R	C	R	C
Athan platform design	I	A	C	R	R	I	R	C
Signed data release	I	C	A	C	I	R	R	C
Privacy and threat model	I	C	C	R	C	C	C	A R
Release readiness	A	R	R	C	R	R	R	C
Incident response	I	A	R	R	R	R	R	R

R means Responsible, A means Accountable, C means Consulted and I means Informed.

31.1 Required forums

- Weekly product and delivery review for scope, dependencies and user evidence.
- Weekly accuracy review for discrepancy triage and Country Profile changes.
- Architecture decision review for domain contracts, native boundaries and update mechanisms.
- Release go or no-go with product, accuracy, QA, mobile, security and operations evidence.
- Incident review for any incorrect published time, missed alarm pattern, compromised package or privacy issue.

32 Risks and mitigations

Risk	Impact	Likelihood	Mitigation	Owner
Authority conventions are incomplete or change without structured notice	High	High	Evidence catalog, monitored sources, profile effective dates and regional rollback	Accuracy lead
Users interpret calculated mosque entry as actual mosque Adhan	High	Medium	Distinct labels, information architecture and usability testing	Product
Mobile OS suppresses or changes alarm behavior	High	High	Native adapters, capability states, device lab, Athan Test and rapid compatibility releases	Mobile lead
High-latitude results appear arbitrary	High	Medium	Profile-controlled method, visible disclosure and polar benchmark suite	Accuracy lead
Time-zone law changes after app release	High	Medium	TZDB monitoring, priority signed packages and targeted stale warning	Data operations
Open mosque data has duplicates or poor coordinates	Medium	High	Precision metadata, deduplication, moderation and optional feature boundary	Data lead
Iqama estimate is mistaken for verified fact	High	Medium	Separate data types, ranges, confidence and prohibited wording tests	Product and QA

Risk	Impact	Likelihood	Mitigation	Owner
Open-source dependency becomes vulnerable or abandoned	Medium	Medium	SBOM, update SLA, replaceable adapter and internal ownership of critical algorithms	Architect
Location telemetry erodes trust	High	Low	Local processing, no advertising ID, opt-in redacted diagnostics and privacy tests	Security
Scope expands into a generic content application	Medium	Medium	Release principles, outcome scorecard and roadmap change control	Product
Profile update changes already scheduled alarms unexpectedly	High	Medium	Generation IDs, future-only reconciliation, user-visible change note and rollback	Mobile and data
Small team cannot cover global authority interpretation	High	High	Phased country rollout, explicit unsupported state and repeatable evidence workflow	Business

33 Decisions and next approvals

Decision	Recommendation	Approval needed by
Launch countries	Limit MVP to a small set with strong authority evidence and meaningful geographic variation	End of Sprint 0
Minimum OS versions	Set after AlarmKit and Android exact-alarm feasibility results	End of Sprint 1
Android exact-alarm permission	Choose SCHEDULE_EXACT_ALARM or eligible USE_EXACT_ALARM after store-policy review	Before Sprint 6
Country Profile governance	Two-person domain review and signed evidence report	Before first beta
Mosque dataset	Treat as optional regional package and complete ODbL review	Before Sprint 11
Verified Iqama	Require traceable source, freshness and revocation path; otherwise show Estimated	Before Sprint 13
Telemetry	Default to coarse anonymous counts; diagnostic bundle opt in	Before MVP beta
External review	Commission independent security, privacy and numerical review before V2	During Sprint 16

With these decisions, UX can begin the home, Accuracy Passport, Athan Health, permission, travel and mosque journeys while architecture and engineering build the calculation contracts and native alarm spikes. The launch-country choice is the only business input that materially changes the first profile backlog.

Appendix A Country Profile example

The following abbreviated example shows the intended structure. Values are illustrative and must not be promoted without authority evidence and review.

```
profileId: country.authority.variant
schemaVersion: 1
version: 2026.09.1
effectiveFrom: 2026-09-01
jurisdiction: XX
authority:
  name: Example Authority
rules:
  fajr: { type: solarAltitude, degrees: -18.0 }
  isha: { type: solarAltitude, degrees: -17.0 }
  asr: { shadowFactor: 1 }
  highLatitude: { trigger: eventMissing, method: anglePortion }
adjustments:
  dhuhrMinutes: 1
rounding: nearestMinute
evidence:
  benchmarkSet: xx-2026-q3
compatibility:
  minimumEngineSchema: 1
signature:
  algorithm: Ed25519
  keyId: profiles-2026-01
```

Appendix B Calculation trace example

Field	Example
Calculation ID	GS 20260924 ABC123
Observer	25.2048 N, 55.2708 E; horizontal accuracy 12 m
Elevation	18 m orthometric; DEM package srtm regional 2026.1
Civil time	Asia Dubai; TZDB 2026a
Engine	Geo Salah 1.0.0; solver tolerance 0.25 s
Country Profile	UAE authority variant 2026.09.1
Raw event	Fajr root at solar altitude configured by profile
Adjustments	Authority offset 0 min; rounding nearest minute
Result	Local displayed minute plus UTC instant retained internally
Confidence	Calculated; authority alignment status from benchmark report

Appendix C Definition of done

- Acceptance criteria pass on supported iOS and Android physical devices where the feature is platform relevant.
- Automated unit, conformance, integration, security and accessibility tests pass with no unexplained benchmark regression.
- Offline and stale-data behavior has been tested and the user receives accurate status language.
- Analytics and logs contain no raw location or prohibited identifiers.

- Country Profile or algorithm changes include evidence, versioning, reviewer approval and rollback instructions.
- User-facing copy distinguishes Calculated, Authority aligned, Verified, Estimated and Unknown consistently.
- Architecture records, API contracts, support guidance and release notes are updated.
- No Severity 1 or Severity 2 defect remains open for calculation, alarm delivery, privacy or package integrity.

Appendix D Glossary

Term	Meaning
Adhan or Athan	The call to prayer; Athan is used in feature names where established
Authority aligned	A calculated result demonstrated to match the approved authority convention within a declared tolerance
Country Profile	Versioned policy defining prayer rules and publication conventions for a jurisdiction
DEM	Digital Elevation Model used when device altitude is unavailable or unreliable
Geo Salah Engine	The local calculation system combining astronomy, prayer rules, civil time and policy
Iqama	The start of the congregational prayer, distinct from prayer entry and Adhan
Provenance	The inputs, versions and adjustments that explain a result
SPA	Solar Position Algorithm; a published high-precision solar-position reference
TZDB	IANA Time Zone Database containing civil-time rules and transitions
Verified	Data supported by an approved traceable source with current integrity metadata

Appendix E Primary technical references

1. National Renewable Energy Laboratory. Solar Position Algorithm for Solar Radiation Applications. https://www.nrel.gov/docs/fy08osti/34302.pdf
2. IANA. Time Zone Database and release information. https://www.iana.org/time-zones
3. Apple Developer Documentation. Scheduling an alarm with AlarmKit. https://developer.apple.com/documentation/AlarmKit/scheduling-an-alarm-with-alarmkit
4. Android Developers. Schedule alarms. https://developer.android.com/develop/background-work/services/alarms
5. NASA Earthdata. Shuttle Radar Topography Mission products and spatial resolution. https://www.earthdata.nasa.gov/centers/lp-daac
6. OpenStreetMap Foundation. Copyright and license. https://www.openstreetmap.org/copyright
7. Flutter Documentation. Flutter SDK overview and platform interoperability. https://docs.flutter.dev/tools/sdk
8. Flutter Documentation. Continuous delivery with Flutter. https://docs.flutter.dev/deployment/cd

References establish the technical basis for the architecture. Country-specific religious and authority sources must be added to each Country Profile evidence pack before that profile can be approved for production.